

MotIQ - Intelligent Motor Health & Power Monitoring System

Electrical, Thermal, and Operational Insight in One Device

Product Overview

Induction motors rarely fail suddenly. Electrical imbalance, rotor defects, deteriorating electrical connections, and thermal hotspots typically develop long before conventional protection devices trip. MotIQ is an intelligent motor health and power monitoring system designed to identify these early warning signs and enable true predictive maintenance.

The system combines Motor Current Signature Analysis (CSA), three-phase power and energy monitoring, infrared thermal analysis, local visualization, and industrial communication in a single compact device suitable for continuous operation in industrial environments.

Product Models

MotIQ is an intelligent motor health and power monitoring system is available in three standard models. All models share the same core monitoring platform, while additional capabilities are enabled based on the selected model.

Pro Model:

The Pro model provides electrical and mechanical motor health monitoring using CSA, along with full power and power quality analysis, local visualization, and BMS or SCADA integration. Thermal monitoring and motor control or automation functions are not included.

Ultra Model:

The Ultra model includes all Pro model features and adds infrared thermal monitoring of contactors and electrical connections. Motor control and automation functions are not included.

Premier Model:

The Premier model includes all Ultra model features and additionally supports motor control and automation functions such as manual motor ON and OFF from the local display, remote dry contact control, and automation or interlock readiness.

Electrical and Mechanical Health Monitoring (CSA)

Using Motor Current Signature Analysis, the system continuously evaluates three-phase motor currents together with a voltage reference to identify electrical and mechanical abnormalities. CSA enables effective condition monitoring without vibration sensors and operates reliably even in harsh or inaccessible locations.

CSA-based motor health indicators include:

- Broken rotor bar detection
- Rotor eccentricity (static and dynamic)
- Electrical unbalance
- Load unbalance
- Slip frequency estimation
- Torque hunting and load oscillation
- Abnormal load variation
- Indicative stator electrical anomalies
- Possible misalignment (CSA-indicative)
- Possible bearing-related electrical signatures
- Long-term CSA trend monitoring

Power and Energy Analysis

The system continuously measures and trends electrical power parameters to support energy optimization, reliability improvement, and early fault detection. Power analysis complements CSA by providing visibility into the interaction between the motor and the electrical supply.

Measured parameters include RMS voltage and three-phase current, active, reactive, and apparent power, power factor, frequency stability, voltage and current harmonic distortion, running hours, and energy consumption. These measurements enable both motor health assessment and electrical performance monitoring.

Thermal Monitoring of Contactors and Connections

An integrated infrared camera continuously monitors critical electrical components inside the motor control center. Thermal monitoring allows early identification of loose connections, contact degradation, oxidation, and high-resistance joints before they lead to catastrophic failures or fire risks.

Thermal monitoring capabilities include an 8 by 8 thermal data array, contactor temperature monitoring, connection hotspot detection, uneven phase heating detection, high temperature alarms, and long-term thermal trend analysis.

Motor Control and Automation

In the Premier model, the system also supports motor control and automation functions. These include manual motor ON and OFF commands from the local display, remote dry contact control, automation readiness, and operation under BMS or SCADA supervision.

Local Visualization and System Integration

An industrial-grade LCD with 800 by 480 resolution provides real-time data, alarms, thresholds, and health indicators directly at the panel.

For system-level integration, the device communicates using Modbus RTU over RS485, with optional BACnet TCP/IP support. Only processed health indicators and alarms are transmitted, reducing network load while enabling seamless integration with BMS, SCADA, PLC, and energy management systems.

Customization and Compliance

Each system is configured according to the motor nameplate data to ensure accurate measurement and reliable diagnostics. For motors rated above 75 kW, the device is supplied with appropriately rated current transformers and measurement ranges.

Key Benefits

The system enables early fault detection, reduces unplanned downtime, lowers maintenance costs, extends motor and contactor lifetime, and improves overall electrical system reliability, safety, and energy performance.

Typical Applications

MotIQ is an intelligent motor health and power monitoring system is suitable for a wide range of industrial and commercial applications, including pumps and booster stations, HVAC fans and air handling units, compressors, conveyors, critical motor control center feeders, and facilities in water, wastewater, oil and gas, and manufacturing sectors.

Disclaimer: Diagnostic information is provided to support condition-based maintenance using electrical and thermal analysis and should be interpreted using engineering judgment. This product does not replace required motor protection or safety devices.